ITI123 Generative AI & Deep Learning Final Presentation

Chia Rui Qi, Cherie
8708815V

Objective

To predict the engagement performance of Instagram posts using historical post data

By predicting engagement levels in advance, businesses can assess their own content or influencer posts, and:

- make more informed decisions in content planning and marketing campaigns
- reduce risk and time spent
- improve return on investment

Dataset

2 datasets were considered for this project, and they both provide the following features originally:

- Small dataset: 1968 posts
- Large dataset: 100k posts randomly selected from 10M posts

user_id/username

followers

following

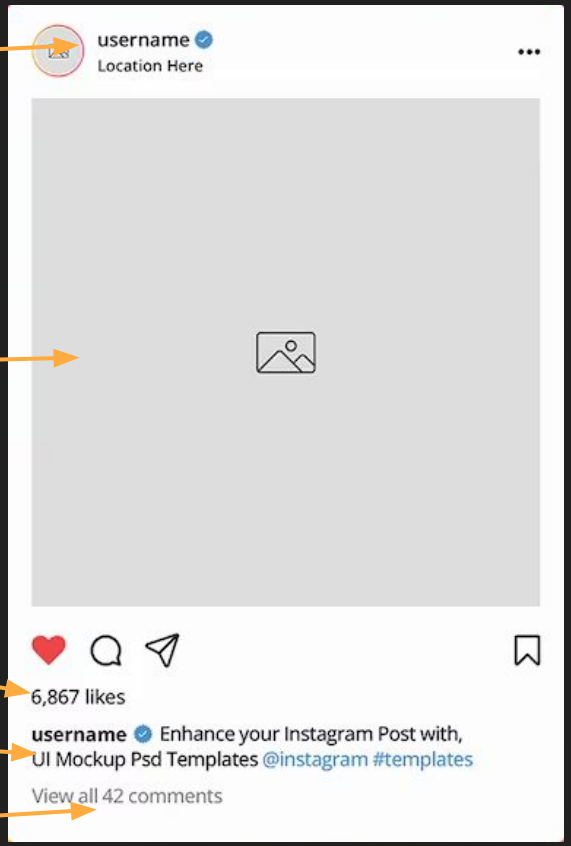
image_path

(1st image only)

likes

caption

comments



Other metadata:

has_location

is_carousel

num_images

is_sponsored

publish_timestamp

Engineered Dataset

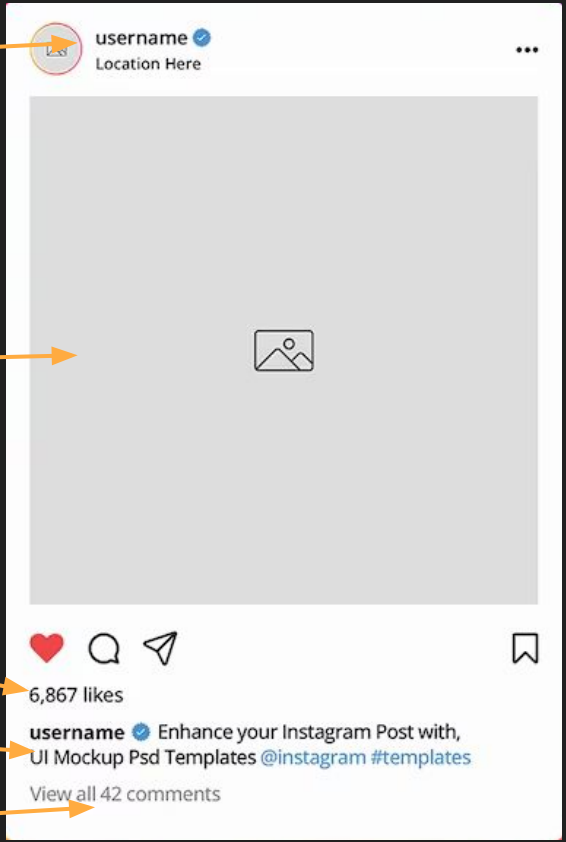
~~user_id/username~~

~~followers~~

~~following~~

follower_following_ratio

image_path



likes

caption

comments

caption_word_count

~~hashtags~~

num_hashtags

Other metadata:

has_location

is_carousel

num_images

is_sponsored

~~publish_timestamp~~

hour

day

is_weekend

Output: Engagement labels

Engagement rate = (likes + comments) / followers

Engagement label: Split into 3 balanced classes based on count in each class (low, moderate, high engagement rate)

Label	Small dataset (1968 posts)	Large dataset (100k posts)
Low	< 0.006%	< 0.02%
Moderate	0.006% to 0.02%	0.02% to 0.04%
High	> 0.02%	>0.04%

Models

following
follower_following_ratio
num_images
has_location
is_sponsored
is_carousel
caption_word_count
likes
hour
day
comments
is_weekend
num_hashtags
caption
image_path

MLP

Preprocessing: One-Hot encoding, StandardScaler
2x of FC (64/32) → ReLU → Dropout → Output logits (3)

LSTM

Preprocessing: Tokenize captions (by words in train & test dataset. Vocab built only from train dataset.)
Token IDs → Embedding layer (50) → LSTM (sequential) → Final hidden memory state (64) → Dropout
→ FC (32) → ReLU → Dropout → Output logits (3)

BERT

Preprocessing: Tokenize captions using BertTokenizer (by subwords in train & test dataset. Vocab of 30,522 tokens.)
Token IDs + attention mask → Pretrained BERT
→ CLS pooling (768) → Dropout
→ FC (256) → ReLU → Dropout → Output logits (3)
Unfreezing: last 2 encoder layers

ResNet18

Preprocessing: Image transforms (resize, random horizontal flip, normalize)
Image → Pretrained ResNet-18 backbone (512) → Dropout
→ FC (128) → ReLU → Dropout → Output logits (3)
Unfreezing: layer4 last residual block group

Hyperparameters

Hyperparameters were tuned using WandB Sweeps with the Bayes method.

Parameters	MLP	LSTM	BERT (frozen)	BERT (unfrozen)	ResNet18 (frozen)	ResNet18 (unfrozen)
hidden_dim	Hd1: [32, 64, 128] Hd2: [16, 32, 64]	[32, 64, 128]	256	256	128	128
embed_dim		[50, 100]				
dropout	min: 0.3, max: 0.5	min: 0.3, max: 0.6	min: 0.3, max: 0.5	min: 0.3, max: 0.5	min: 0.3, max: 0.5	min: 0.3, max: 0.5
max_len		[32, 64, 128]	[64, 128]	[64, 128]		
learning_rate	min: 0.0005, max: 0.001	min: 0.0005, max: 0.001	min: 0.0005, max: 0.001	min: 0.00002, max: 0.00005	lr_head min: 0.0005, max: 0.002	lr_backbone: min: 0.0001, max: 0.001, lr_head min: 0.0005, max: 0.002

Fusion Model

following
follower_following_ratio
num_images
has_location
is_sponsored
is_carousel
caption_word_count
likes
hour
day
comments
is_weekend
num_hashtags
caption
image_path

MLP

LR: 0.000915
Dropout: 0.443
HD1: 128, HD2: 64

BERT

LR: 0.000529
Dropout: 0.337
Max_len: 128

ResNet18

Lr_backbone: 0.000253
Lr_head: 0.00161
Dropout: 0.370

Take features from all models right before their classifier heads:
32 (MLP) + 256 (BERT) + 128 (ResNet18) = 416

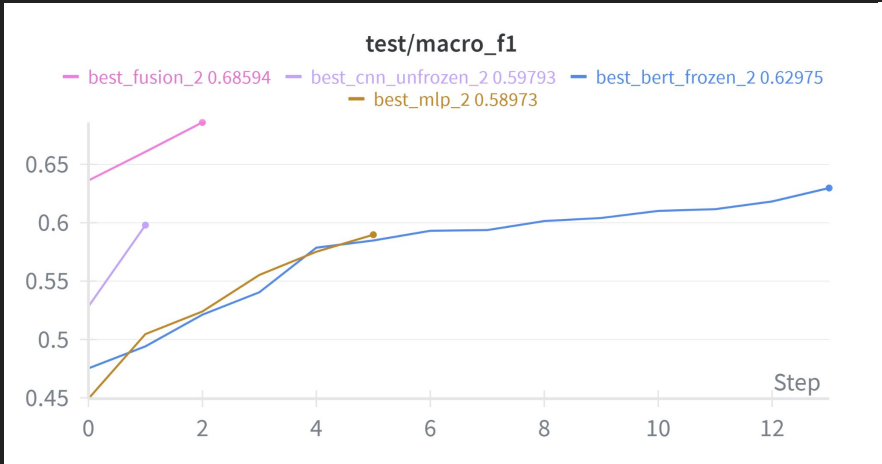
Fusion

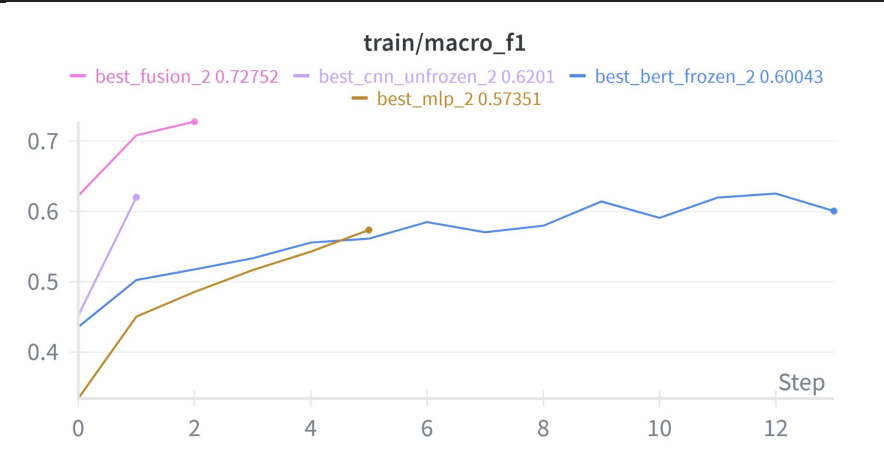
(optional: FC → Sigmoid)
→ FC (256) → ReLU → Dropout
(optional: → FC (256) → ReLU → Dropout)
→ Output logits (3)

LR: 0.000675
Dropout: 0.497
HD: 128

All the best models, and fusion

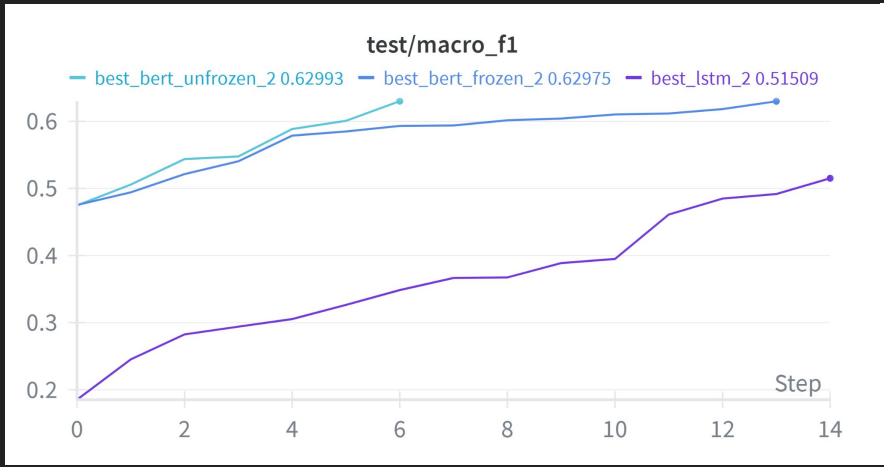
- The best MLP has macro-F1 score of 59%
- The overall Fusion Model has best macro-F1 score of 68.6%

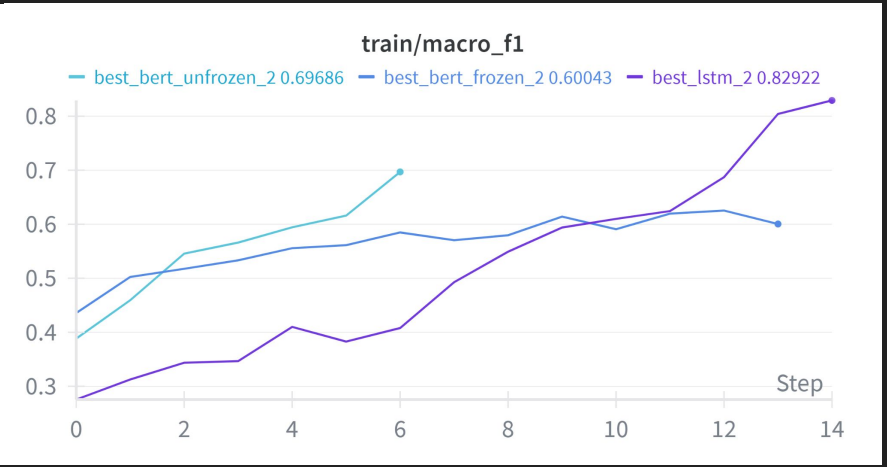




- Mild overfitting in fusion model is observed from the drop in macro-F1 from train to test (72% to 68%)
- The fusion model did help to improve the macro-F1 score, but the final performance is still only moderate.

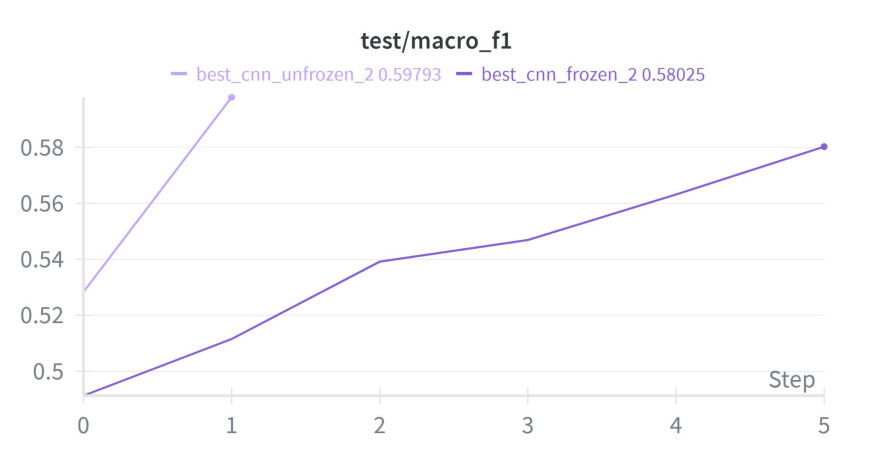
LSTM vs Frozen BERT vs Unfrozen BERT

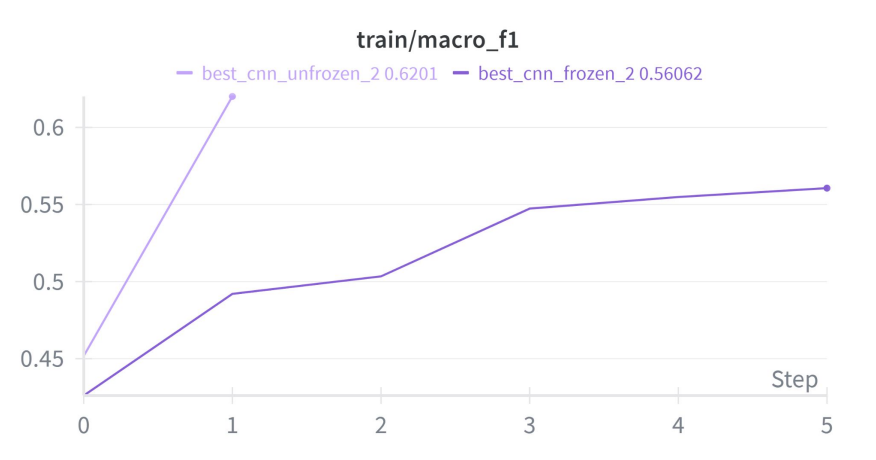




- After sweeps, LSTM overfit more and performed worse than frozen BERT, leading to the selection of BERT for captions.
- Frozen and unfrozen BERT had similar test macro F1
- Unfrozen BERT overfit (train 69.7%, test 63%), while frozen BERT achieved comparable test performance (train 60%, test 63%) suggesting stable generalization.
- Frozen BERT is the final model for captions, with best macro-F1 score of 63%

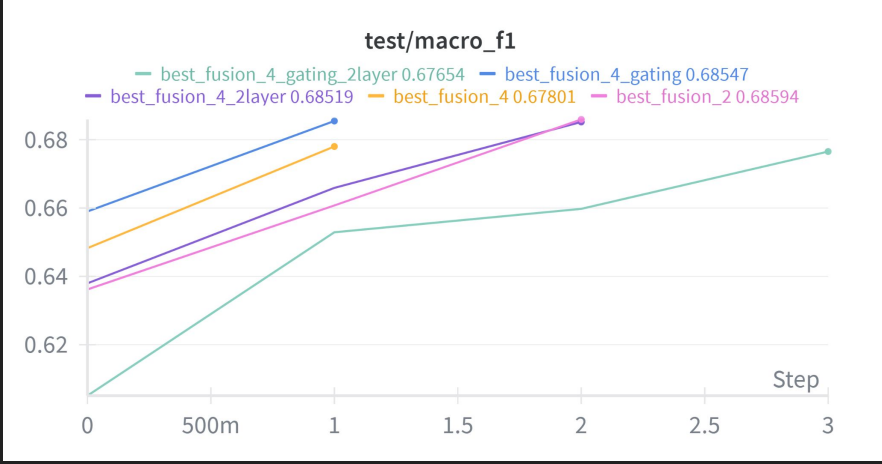
Frozen ResNet18 vs Unfrozen ResNet18





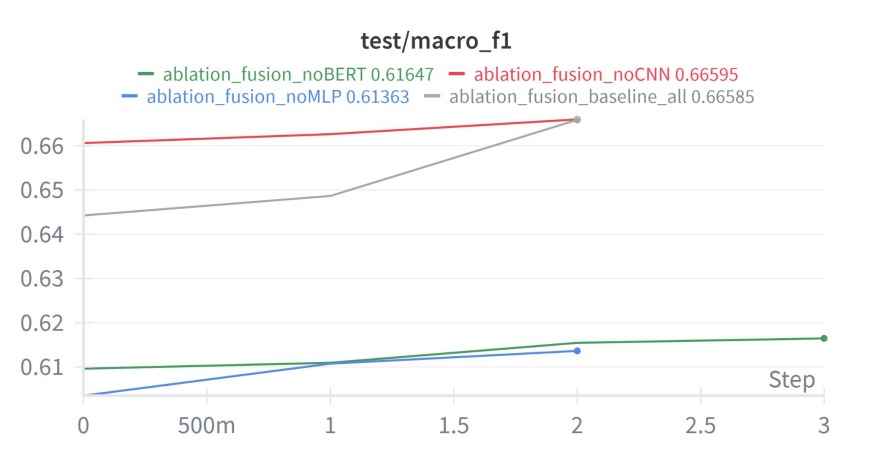
- In both test and train, the unfrozen ResNet18 performed better
- There was little to no overfitting on ResNet18, whether frozen or not
- Unfrozen ResNet18 had little to no overfitting (train 62%, test 59.8%), while frozen ResNet18 achieved comparable test performance (train 56%, test 58%), suggesting stable generalization.
- Unfrozen ResNet18 is the final model for images, with best macro-F1 score of 59.8%

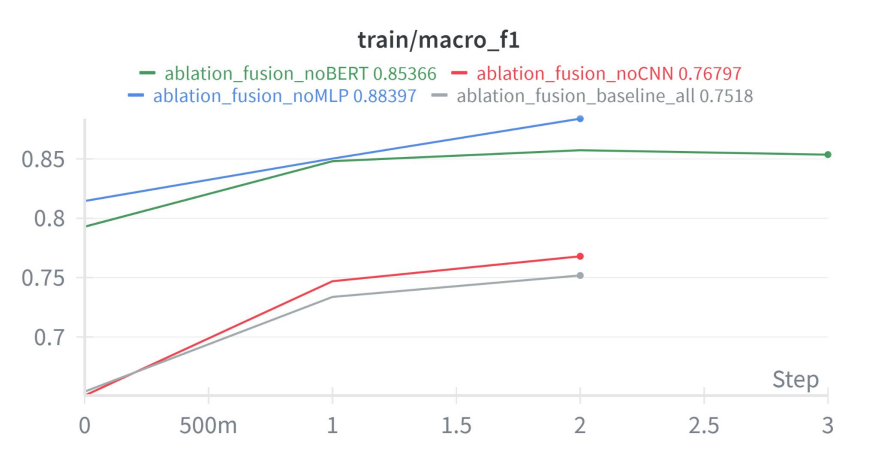
Comparing Fusion Models



- The basic 1-layer fusion model performed the best at inference time
- It seemed like the basic 1-layer fusion model overfit the most when rerun, but this could be attributed to randomness.
- Having either the 2nd layer or gating in the fusion model gave a similar performance as the 1-layer fusion model and did not help, likely because of a dataset size limitation, or that models already produce high level embeddings, or because features went through a bottleneck layer before fusion.
- Having both the 2nd layer & gating in the fusion model worsened the performance slightly

Ablation study

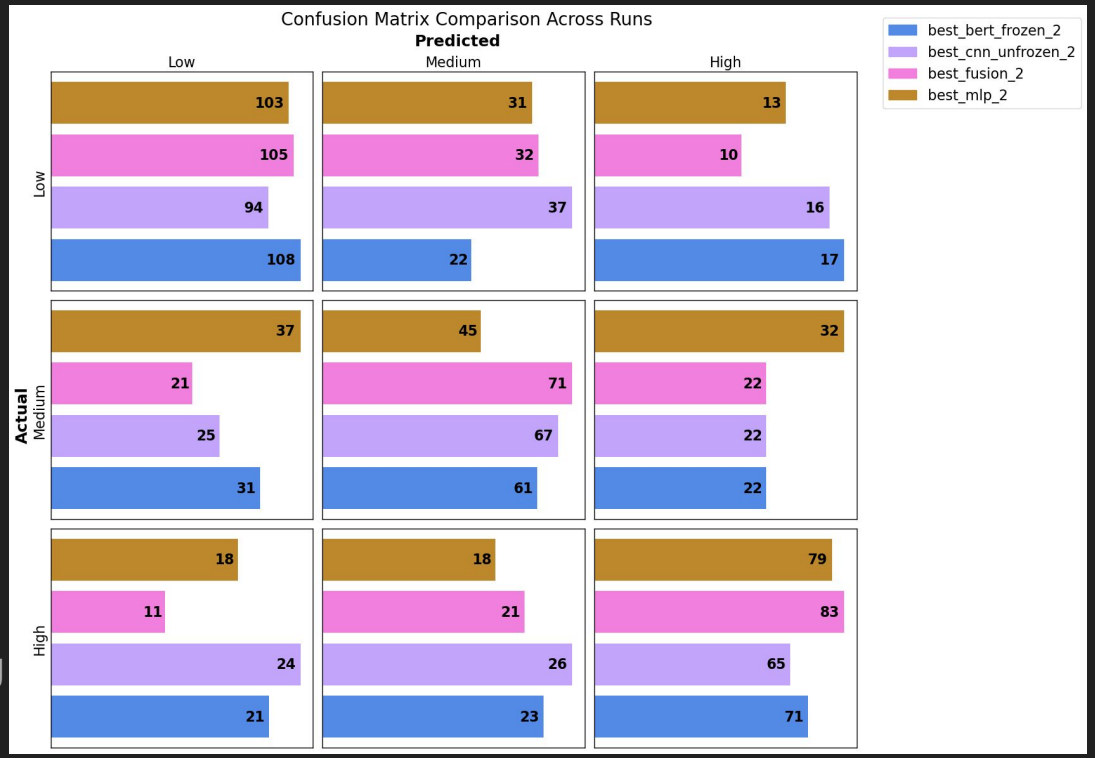




- This is to satisfy my curiosity on which models - BERT, ResNet18 or MLP - had the most positive or negative influence on the fused model outcome.
- Interestingly, the fusion model performed the best when it fused all models, or only excluded ResNet18. This means that ResNet18 had almost no influence on predicting engagement labels, and BERT and MLP had similar contribution.

Conclusions

- Individual models achieved a macro-F1 of 59% to 63%, significantly better than the 33% random-guess baseline, indicating borderline moderate performance.
- BERT and captions were slightly more effective.
- The fusion model reached 68% macro-F1, confirming that combining features improves engagement label prediction, though its performance is only decent.



- Many samples with Medium/Moderate engagement were predicted as Low, and vice versa.
- The Medium/Moderate engagement label also has the fewest correct predictions.

Possible improvement

- Larger dataset with wider spread of engagement rates
- Labelling if engagement rate was higher than a certain threshold, instead of 3 classes
- Add in use of hashtags that are currently unused
- More intermediate FC layers before classifier head for more gradual bottleneck
- Different way to merge the models instead of just FC layers: tried 2 FC layers, tried gating.
- Fuse features earlier, before the bottleneck layer (pretrained model → dropout → fuse)
- Using multimodal transformers instead, for cross-modal relationships to be learned through multiple layers
- Widen hyperparameter sweep boundaries